

Identifying abrupt changes and detecting gradual trends of annual rainfall in an Indian arid region under heightened rainfall rise regime

Deepesh Machiwal,^{a*} Sanjay Kumar,^b Devi Dayal^a and Shamsudheen Mangalassery^a

^a Regional Research Station, ICAR-Central Arid Zone Research Institute, Bhuj, Gujarat, India

^b Krishi Vigyan Kendra, ICAR-CAZRI, Bhuj, Gujarat, India

ABSTRACT: This study aimed at identification of abrupt change points (CPs) and detection of gradual trends in 34-year (1980–2013) annual rainfall at nine stations of an Indian arid region. The CPs were determined by five tests and their significance was examined by two tests. Furthermore, trends were tested by three tests and their magnitudes were quantified by two tests. Novelty of the study lies in investigating significance of trends sequentially over years by applying Mann–Kendall (M–K) test. The identified CPs were similar for standard normal homogeneity test and cumulative deviations test at most stations. In contrast, Pettitt and Bayesian tests detected CP in years 2002 and 2005 at six and three stations, respectively, and their significance was verified. Results of sequential M–K test did not match with other tests' results. The mean annual rainfall after CP (350–627 mm) increased by 14–80% of the amount before CP (306–444 mm) with 7–42% reduction in coefficient of variation. The box–whisker plots supported these findings. Results of trend tests indicated statistically significant trends at Anjar, Bhachau, Mandvi and Rapar. Trend magnitudes by linear regression prior to CP (–6.2 to 7.1 mm year^{–1}) showed an overall increase after CP (4.7–40.8 mm year^{–1}) with negative trend at one station. Sen's slope test, showing good harmony with linear regression, revealed that trend magnitudes after CP were 2–10 times higher than that for 34-year period at six stations. Results of M–K test applied for sequential periods emphasized that rainfall trends are becoming stronger over time. This finding suggests that significance level of increasing rainfall trends may further increase in future. Finally, findings of this study are useful for planners and decision makers for developing policies to meet the challenges of the heightened rainfall in study area. Also, the approach used here may be adopted in other parts of the globe.

KEY WORDS abrupt change; arid region; gradual trend; multiple statistical tests; rainfall series; variability

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1. Introduction

Rainfall is one of the important hydrological variables to diagnose the impact of climate change and variability on the availability of the water resources in a region (Cannarozzo *et al.*, 2006). The spatial and temporal patterns of the rainfall affect the eco-hydrological processes, which control and regulate the evolution of surface ecosystem (Oguntunde *et al.*, 2006; Liu *et al.*, 2008). In arid and semi-arid regions, abrupt changes (or shifts) and gradual changes (or trends) causing the rainfall variability over the temporal and spatial scales have profound influence on the dynamic linkages between ecosystems and water cycle mainly due to the limiting water resources, which is constrained not only for scarce quantities but also for its intermittency and unpredictable nature (Rodriguez-Iturbe and Porporato, 2005). Thus, the shifts and trends in the rainfall series are important from both the scientific and practical point of views (Tosic, 2004; Tabari and Talaei, 2011).

Many studies performed rainfall variability analyses all over the world with an increasing interest in analysing trends and abrupt changes (Mirza *et al.*, 1998; Pugacheva *et al.*, 2003; Astel *et al.*, 2004; Machiwal and Jha, 2008; Kumar *et al.*, 2010; Deka *et al.*, 2013; Goyal, 2014; Talaei, 2014).

It is realized that such studies are mostly carried out in humid and semi-arid regions; however, and the studies for arid regions are rare, e.g. Ouarda *et al.* (2014); Machiwal *et al.* (2016). The importance of spatial and temporal investigations of the rainfall for arid regions is highlighted by Machiwal *et al.* (2016). Hess *et al.* (1995) examined presence of trends in annual rainfalls and in number of rainy days in the arid northeast Nigeria, and results revealed the significantly decreasing trends in both the variables. Tabari and Talaei (2011) analysed trends in annual and seasonal rainfall at 41 sites in arid and semi-arid areas of Iran by using Mann–Kendall (M–K), linear regression and Sen's slope tests. The results indicated decreasing trends of annual rainfall at 60% sites and mostly decreasing trends in rainfall of spring and winter seasons. Ouarda *et al.* (2014) evaluated significance of trends and change points (CPs) in total annual and monthly

* Correspondence to: D. Machiwal, Regional Research Station, ICAR-Central Arid Zone Research Institute, Kukma, Bhuj, Gujarat 370105, India. E-mail: dmachiwal@rediffmail.com

rainfall, annual and monthly maximum daily rainfall and number of rainy days per year and per month at four sites located in arid region of United Arab Emirates. The results showed non-significant decreasing rainfall trends in most of the annual series and strongly decreasing trends in rainfall of February and March months. The CP in all the rainfall series was revealed in year 1999 at all sites. Studies involving use of global and regional climate models also indicate a likely decrease in rainfall quantities in most of the arid and semi-arid regions across the globe, although large variability is seen in the predicted scenarios for arid areas (Black *et al.*, 2010; Hemming *et al.*, 2010; Chenoweth *et al.*, 2011).

The arid lands extend over 61 million km² on the globe (46% area worldwide) (FAO-AGL, 2003) and experience a general water deficit scenario mainly due to less frequency and low magnitudes of the rainfall. In India, arid lands are spreading over 38.7×10^6 ha and experience both hot and cold climates (Kar *et al.*, 2009). The hot Indian arid zone, occupying about 32×10^6 ha, mainly falls in the western Rajasthan (62% of hot arid lands) and in the northern Gujarat (19.6% of total arid lands) (Kar *et al.*, 2009). It is revealed that the occurrences of heavy to very heavy rainfall events are increasing over the Indian subcontinent, whereas the low and moderate rainfall events are decreasing (Goswami *et al.*, 2006; Ghosh *et al.*, 2009). Moreover, the occurrence of very heavy rainfall over a short period of time and continuous occurrence of dry hours produce adverse effect on crop, while the continuous occurrence of low rainfall intensity favours the crop growth (Revadekar and Preethi, 2010). The hourly rainfall intensity is high over the Gujarat region in all the time (Varikoden *et al.*, 2012). It indicates that high intensity regions are receiving rainfall from very few events, namely, these regions receive considerable amount of rainfall within a short period of time. In Gujarat, the rainfall intensity is recorded to be the maximum (>5.5 mm/rainy hours) at almost all the hours of a day (Varikoden *et al.*, 2012). Besides these facts, it is reported that the rain shadow regions of the northern parts of Gujarat especially Kachchh and its surrounding areas showed rainfall maxima during night hours, around 1500–1900 UTC (Varikoden *et al.*, 2012). Kachchh, the largest district of Gujarat State situated in the northern part and the second largest district of the country, experiences hot arid climate over the entire land (Harsh and Tewari, 2007; Dayal *et al.*, 2009). In Kachchh, a considerable portion of the annual rainfall ranging from 38 to 68% occurred in consecutive 2–4 days during the 7-year (2007–2013) period (Machiwal *et al.*, 2015). Thus, there is a need to understand the spatial and temporal changes and trends of the rainfall in this hot arid region. Therefore, this study is performed with the objective of identifying abrupt changes, verifying their significance, detecting gradual trends and quantifying their magnitudes in the annual rainfall of Kachchh by employing multiple statistical tests. Furthermore, this study involves a new approach to examine the significance of the gradual trends sequentially over the years by applying the M–K trend test.

2. Materials and methods

2.1. Study area and climate

Kachchh district (study area) of Gujarat encompasses 45 612 km² area, and is located between 22°44'08" and 24°41'30"N latitudes, and 68°07'23" and 71°46'45"E longitude (Figure 1). It is almost like an island situated in the westernmost portion of the country having land connectivity only through roads and railway tracks. The study area is surrounded by the waterbodies with presence of the Arabian Sea in west, the Gulf of Kachchh in south, the Great Rann in north and the Little Rann in east directions. Both, the Great and Little Ranns represent the salty marshlands. The district is divided into ten administrative blocks, i.e. Bhuj and Nakhatrana (north), Lakhpat and Abdasa (west), Mandvi, Mundra, Anjar and Gandhidham (south), and Bhachau and Rapar (east).

The climate of the area is characterized as tropical and arid, experiencing extreme weather conditions of long hot summers and cold winters with very high aridity index (more than 40%) having more or less dry desert conditions (Prusty, 2012). The area receives a large portion of the scanty annual rainfall from the southwest monsoon during June to August months. The 100-year mean annual rainfall is 341 mm with highly erratic temporal pattern (source: http://indiawaterportal.org/met_data/). The mean monthly air temperature is generally the highest in May ranging from 22.1 to 31.9 °C, and the lowest in January varying from 8.8 to 22.7 °C from the 100-year mean values (source: http://indiawaterportal.org/met_data/). Relative humidity is more than 80% in the coastal areas and more than 65% in inland during the monsoon season. In rest of the year, the air is generally dry and the relative humidity in afternoon falls below 25%. The potential evapotranspiration (PET) varies from 1750 mm year⁻¹ in the coastal area of Naliya, Mandvi and Mundra to 1900 mm year⁻¹ towards Bhuj and Anjar. Towards the northern and northeast portions, the PET values decrease to 1800 mm year⁻¹ and less (Singh and Kar, 1996).

2.2. Data description

Monthly rainfall data were collected for ten rainfall gauging stations of the area. Each administrative block has one rainfall gauging station located in the block headquarter; however, the raingauges of Abdasa and Lakhpat blocks are situated at Naliya and Dayapar, respectively. Location of the ten rainfall gauging stations is shown in Figure 1. The rainfall data of 34-year (1980–2013) period were collected from the State Emergency Operation Centre and State Flood Control Room, Revenue Department, Gandhinagar, Gujarat, India. The raingauge in the Gandhidham block was established in the year 1997–1998 when this portion of the land was established as a new block in 1997, and thus, the rainfall data for this block could be available only for a period of 16 years (1998–2013). Therefore, this study utilized the rainfall data monitored at nine rainfall gauging stations only without analysing the rainfall data of Gandhidham block. All the collected rainfall data were

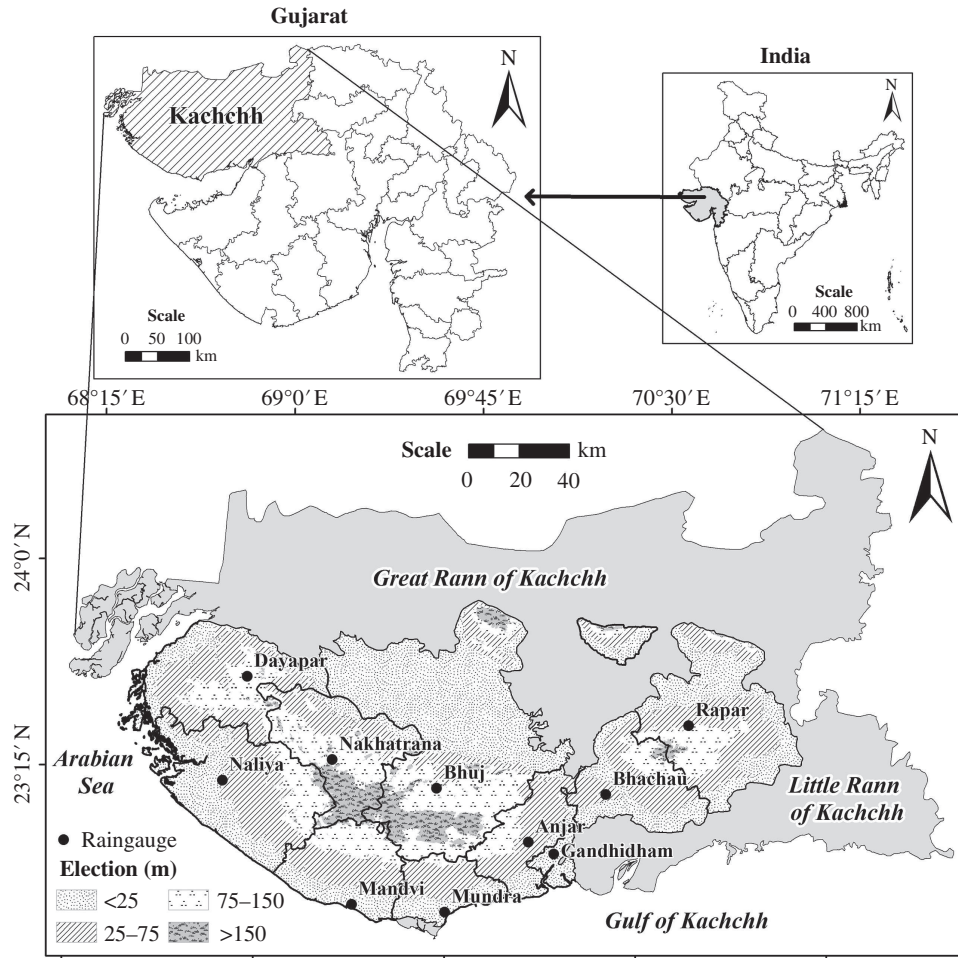


Figure 1. Location map of study area.

checked for consistency and regularity, and were subsequently used for the analyses.

2.3. Identifying abrupt changes in rainfall time series

In a hydrological time series, an abrupt CP (or shift period) is usually not known, and under this condition, the most-widely used statistical tests to identify the unknown CPs are standard normal homogeneity (SNH) test, Pettitt test, cumulative deviations (CD) test, Bayesian test and sequential M–K test. These five tests were used in this study to detect the abrupt CPs in annual rainfall of nine gauging stations during the period of study (34 years), where initially the CPs were unknown. The approach of adopting multiple tests, both statistical (first four tests) and graphical (last test), is recommended in the hydrologic time series analysis (e.g. Machiwal and Jha, 2008; Sonali and Nagesh Kumar, 2013; Machiwal and Jha, 2016).

The SNH test statistic (Alexandersson, 1986) applies to a time series of size n having zero mean and unit variance. The test statistic is given as follows (Alexandersson and Moberg, 1997):

$$T_{\text{SNH}}^2 = \max_{1 \leq a \leq n-1} \{T_a^2\} = \max_{1 \leq a \leq n-1} \left\{ a \cdot \bar{z}_1^2 + (n-a) \cdot \bar{z}_2^2 \right\} \quad (1)$$

where, a = CP year, $\bar{z}_1$ = mean of the data points from the year 1 to a and $\bar{z}_2$ = mean of the series from the year $(a+1)$ to n .

The Pettitt (1979) test statistic is less sensitive to outliers in the data series and skewed distribution and is more sensitive to the breaks in the middle of the series (Wijngaard *et al.*, 2003). The test statistic is defined as:

$$T_y = 2 \sum_{i=1}^y r_i - y(n+1), \quad y = 1, 2, \dots, n \quad (2)$$

where, r_i = rank of i th data in the series and n = size of the series. The significant CP occurs at the point (in years) where the value of $|T_y|$ is found maximum as follows:

$$T_{\text{CP}} = \max_{1 \leq y \leq n} |T_y| \quad (3)$$

The CD test statistic, defined by the adjusted partial sums or CD from the mean, is expressed as (Buishand, 1982):

$$S_k^* = \sum_{t=1}^k (x_t - \bar{x}), \quad k = 1, 2, \dots, n \quad (4)$$

where, x_t = data of year t and $\bar{x}$ = mean of the series. The promising CP occurs at the point (year) where the value of

$|S_k^*|$ is maximum, defined as follows:

$$S_{CP}^* = \max_{1 \leq k \leq n} |S_k^*| \quad (5)$$

The Bayesian test statistics, based on weighted rescaled partial sums from the mean, is defined as (Buishand, 1982):

$$Z_k^* = \{k(n-k)\}^{-1/2} S_k^*, \quad k = 1, 2, \dots, n \quad (6)$$

The probable change occurs at the point or year where the value of $|Z_k^*|$ is found maximum, shown as follows:

$$Z_{CP}^* = \max_{1 \leq k \leq n} |Z_k^*| \quad (7)$$

In the sequential M–K test (Sneyers, 1990; Partal and Kahya, 2006; Modarres and Sarhadi, 2009), the sequential values x_i and x_j in a series $x_t = (x_1, x_2, x_3, \dots, x_n)$ are compared to find out the number of times (n_j) when $x_j > x_i$. The test statistic is calculated as:

$$t_j = \sum_{i=1}^j n_j \quad (2 \leq j \leq n) \quad (8)$$

The mean, $E(t_j)$ and variance, $\text{var}(t_j)$ of the test statistic (t_j) are given by:

$$E(t_j) = \frac{j(j-1)}{4} \quad (9)$$

$$\text{Var}(t_j) = \frac{j(j-1)(2j+5)}{72} \quad (10)$$

The sequential values of the normalized variable statistic $u(t_j)$ is then calculated as follows:

$$u(t_j) = \frac{t_j - E(t_j)}{\sqrt{\text{var}(t_j)}} \quad (11)$$

The $u(t_j)$ is the forward sequence, and the backward sequence $u'(t_j)$ is obtained by using the same equation but with reversed series. The null hypothesis (H_0) of no step CP is to be rejected if any of the points in the forward sequence, $u(t_j)$ are outside the confidence intervals at level of significance (l.s.), i.e. $\alpha = 0.01$ and 0.05 . This test is frequently used to know the approximate time of occurrence of the CP, as indicated by the intersection of the test statistic curves of the $u(t_j)$ and $u'(t_j)$. The beginning of a step CP is located by an intersection point of $u(t_j)$ and $u'(t_j)$ situated within the confidence interval (Moraes *et al.*, 1998; Zhang *et al.*, 2011).

2.4. Verifying detected CPs by confirmation tests

The significance of the CPs detected by the five tests was confirmed through a comparative evaluation of their results by applying two other statistical tests, i.e. parametric t -test and nonparametric Mann–Whitney (M–W) test. The t -test was applied by dividing the complete rainfall series at the detected CP into two sub-series of size n_1 and n_2 . It is to note that the second sub-series started from the CP. The

t_s test statistic is given as (Snedecor and Cochran, 1980; Machiwal and Jha, 2012):

$$t_s = \frac{[\bar{x}_2 - \bar{x}_1]}{[s \sqrt{1/n_1 + 1/n_2}]} \quad (12)$$

$$\text{and } s = \sqrt{\frac{(n_1 - 1) s_1^2 + (n_2 - 1) s_2^2}{n - 2}} \quad (13)$$

where, $\bar{x}_1$, $\bar{x}_2$, s_1^2 and s_2^2 = estimated means and variances of the first and the second sub-series, respectively. If the computed t_s value exceeded its critical value at ' $n-2$ ' degree of freedom (d.f.) and significance level (α) of 0.01, 0.05 and 0.10, then the null hypothesis (H_0) of no significant CP was rejected.

In the M–W test, another z_t series was generated by arranging the original data (x_t) in ascending order. The test statistic u_c , defining the null hypothesis (H_0) of no significant CP, is expressed as (Snedecor and Cochran, 1980; Machiwal and Jha, 2012):

$$u_c = \frac{\sum_{i=1}^{n_1} R(x_i) - n_1 (n_1 + n_2 + 1) / 2}{[n_1 n_2 (n_1 + n_2 + 1) / 12]^{1/2}} \quad (14)$$

where $R(x_i)$ = rank of x_i data in series z_t . The H_0 is rejected for $u_c >$ critical value at $\alpha = 0.01, 0.05$ and 0.10 .

2.5. Examining impact of CPs on rainfall statistics

In order to understand the impact of the CPs on the annual rainfall at the nine stations, statistical properties such as the mean, standard deviation (SD) and coefficient of variation (CV) of the rainfall data were computed for the complete series of 34 years (1980–2013), and sub-series before and after CPs. Also, box–whisker plots of the annual rainfall were drawn for the complete and partial sub-series before and after the CPs. The box–whisker plot reveals spatio-temporal variations of the rainfall by summarizing five important statistical properties of a time series, i.e. median, 25th and 75th percentile values, outliers and extremes (USEPA, 2006). In this study, the CPs for nine stations were considered in the years those confirmed from the results of the t -test and M–W test.

2.6. Detecting gradual trends

Of the two types of tests, i.e. parametric and nonparametric, for detecting trends or gradual changes in the time series, this study considered three nonparametric tests, i.e. Kendall's rank correlation (KRC), modified M–K, and Spearman rank-order correlation (SROC) tests. All the trend tests were applied to a complete rainfall series of 34 years (1980–2013) and two sub-series before and after the confirmed CPs. The major limitation of the parametric tests is the requirement of normality in the data series, which is generally not found in the most hydrological data series (Machiwal and Jha, 2012). In contrast, the nonparametric tests are widely used for investigating the trends

in the hydrological series all over the world (e.g. Villarini *et al.*, 2011; Guerreiro *et al.*, 2014).

The SROC test is applied on the difference series $d_t = R_{xt} - t$, where R_{xt} is the rank of x_t in the original series arranged in ascending order; for ties, take the mean of the ranks assigned as there is no tie. Then the coefficient of trend (r_s) is computed as follows (McGhee, 1985):

$$r_s = 1 - \frac{6 \sum_{t=1}^n d_t^2}{n(n^2 - 1)} \quad (15)$$

The H_0 of absence of significant trend is to be rejected if the critical values of the Student's t -distribution at d.f. = $n - 2$ and $\alpha = 0.01, 0.05$ and 0.10 are more than the test statistic (t_s) defined as (McGhee, 1985):

$$t_s = r_s \sqrt{\frac{(n-2)}{(1-r_s^2)}} \quad (16)$$

The test statistics (Z_{KRC}) of the KRC test is given by the following expression (Kendall, 1973):

$$Z_{KRC} = \frac{\tau}{\sqrt{\text{var}(\tau)}} \quad (17)$$

$$\tau = \left\{ \frac{4p}{n(n-1)} \right\} - 1 \quad \text{and} \quad (18)$$

$$\text{Var}(\tau) = \left\{ \frac{2(2n+5)}{9n(n-1)} \right\} \quad (19)$$

where, p = number of times $x_j > x_i$ ($j > i$) is counted in all pairs of (x_i, x_j) in the series.

The M–K test statistic S is computed by using Equations (20)–(22) (Salas, 1993):

$$S = \sum_{k=1}^{n-1} \sum_{j=k+1}^n \text{sgn}(x_j - x_k) \quad (20)$$

$$\text{sgn}(x_j - x_k) = \begin{cases} +1 & \text{if } (x_j - x_k) > 0 \\ 0 & \text{if } (x_j - x_k) = 0 \\ -1 & \text{if } (x_j - x_k) < 0 \end{cases} \quad (21)$$

$$\text{Var}(S) = \frac{\left[n(n-1)(2n+5) - \sum_{i=1}^m t_i(t_i-1)(2t_i+5) \right]}{18} \quad (22)$$

where, m = number of tied groups having similar data for a group and t_i = number of data in i th tied group. For $n > 40$, the test statistic is given as follows (Hirsch *et al.*, 1982):

$$z = \frac{(S + m)}{\sqrt{\text{Var}(S)}} \quad (23)$$

where, $m = 1$ for $S < 0$ and $m = -1$ for $S > 0$. The z -value is considered as zero for $S = 0$. The H_0 of no upward/downward trend is to be rejected if the computed z -value is found to be greater than the critical

value obtained from the normal distribution at three l.s. considered in this study, i.e. $\alpha = 0.01, 0.05$ and 0.10 .

In this study, the original M–K test statistic was modified in order to remove the effect of serial correlation on the test by following the approach proposed by Lettenmaier (1976), and the modified M–K test statistic is given as follows:

$$z^* = z \times \sqrt{\frac{n^*}{n}} \quad (24)$$

where, n^*/n is termed as the correction factor computed for the autocorrelation function (r_1) lag-1 autoregressive process from the following expression (Matalas and Langbein, 1962):

$$\frac{n^*}{n} = \frac{1}{1 + 2 \frac{r_1^{n+1} - n r_1^2 + (n-1) r_1}{n(r_1-1)^2}} \quad (25)$$

2.7. Quantifying gradual trends

Trend magnitudes of the annual rainfall series at nine stations were estimated by using one parametric regression test and one nonparametric Sen's slope estimation technique. The parametric tests are found to be more powerful in detecting the trends in comparison to nonparametric tests. In order to apply the regression test, the rainfall data of the two sub-series were plotted over the time scale, and then linear regression line was fitted separately to each sub-series. The trend magnitude was obtained from the slope of the fitted trend lines.

In addition, the magnitudes of the trends were quantified by using Sen's slope test (Sen, 1968; Hirsch *et al.*, 1982) with test statistic given by the slope (β) as follows:

$$\beta k = \text{median} \left(\frac{x_{ik} - x_{jk}}{i - j} \right) \quad \text{for all } i < j \quad (26)$$

where, x_{ik} and x_{jk} = data at time i and j ($j > i$), and k = station. The positive β -value indicates the upward trend whereas the negative value suggests the downward trend.

2.8. Applying M–K test to investigate sequential significance of trends

In this study, the M–K trend test was applied to investigate the significance of the gradual trends over the years sequentially from 21- to 34-year periods. The approach of investigating sequential significance of trends is considered for the first time in this study. In the proposed approach, first a time series of annual rainfall data for a chosen base period before the existence of the CP, i.e. 21 years (1980–2000) considered in this study was selected. Then, the M–K trend test was applied, and the z_1 test statistic value of the test was computed. Then, the rainfall time series needed to extend by adding one subsequent year's ($n = 22$) rainfall data to the base series, and the M–K test was again applied to compute the z_2 test statistic value for the annual rainfall series of 22-year (1980–2001) period. The earlier step was repeatedly executed by including one more subsequent year's rainfall data to the time series until the last data for the last year was included in the time series, and each time the M–K test was applied

to sequentially compute the test statistic values like $z_3, z_4, \dots, z_n$. Finally, the series of the z test statistic values was plotted against the corresponding periods having different number of years starting from $n = 21$ and up to $n = 34$. To investigate the influence of the occurrence of the abrupt change on the significance of the gradual trends, a linear regression line was fitted to the plot of the z test statistics over number of years. The statistical significance of the sequential gradual rainfall trends under the influence of the abrupt change was checked by computing the coefficient of determination (R^2) values.

3. Results and discussion

3.1. Expected CPs in annual rainfall series

The results of the four CP tests, i.e. the SNH, Pettitt, CD and Bayesian tests, identified years of the expected CPs in the annual rainfall series of nine stations, and the same are presented in Table 1. It is clearly seen from Table 1 that the years of CPs identified by the four tests are not exactly similar for the rainfall series of all the stations. Slight deviations in CPs identified by different tests, i.e. the SNH, Pettitt and CD tests, are also found in other studies performed in other parts of India (e.g. Goyal, 2014; Taxak *et al.*, 2014). This outcome explains the need of applying multiple statistical tests for identifying CPs as adopted in this study, and emphasizes utilizing this approach in future studies as well.

The CPs identified by the SNH test at nine stations are the years 1987, 2008 and 2012 at one station each, and the years 2002, 2005 and 2009 at two stations each. However, the CPs identified by the Pettitt test are either the year 2002 at six stations (Anjar, Bhachau, Bhuj, Dayapar, Mundra and Rapar) or the year 2005 at three stations (Naliya, Mandvi and Nakhatrana) (Table 1). The years indicated as the CPs by the CD test are similar to those identified by the Pettitt test for the six stations, i.e. Naliya, Anjar, Bhuj, Mandvi, Mundra and Rapar. However, the CD test resulted the years 1993, 1991 and 2002 as the CPs at stations Bhachau, Dayapar and Nakhatrana, respectively. The CPs resulted from the Bayesian test are found to be exactly similar to those obtained from the SNH test for all the stations.

The results of the sequential M–K test with values of the normalized variable statistic in forward and backward sequence, i.e. $u(t_j)$ and $u'(t_j)$ are presented in Figure 2(a)–(i) for the nine stations. It is seen from Figure 2(a)–(i) that the forward sequence $u(t_j)$ crosses the confidence line at $\alpha = 0.05$ and 0.01 for all the stations at least once, and hence, the H_0 of no step change is to be rejected. This finding suggests presence of the step changes in the annual rainfall series at all the stations. Furthermore, it is observed that both the forward and backward sequences perfectly intersect each other at three stations, i.e. Naliya, Dayapar and Nakhatrana (Figure 2(a), (e) and (h)). At remaining six stations, the $u(t_j)$ and $u'(t_j)$ curves either do not have apparent intersection, i.e. for Anjar, Bhachau and Rapar (Figure 2(b), (c) and (i)) or

Table 1. Years identified as change points by four change point detection tests in annual rainfall of nine rainfall stations of Kachchh district.

Station	Standard normal homogeneity test	Pettitt test	Cumulative deviations test	Bayesian test
Naliya	2008	2005	2005	2008
Anjar	2002	2002	2002	2002
Bhachau	2009	2002	1993	2009
Bhuj	2005	2002	2002	2005
Dayapar	1987	2002	1991	1987
Mandvi	2005	2005	2005	2005
Mundra	2012	2002	2002	2012
Nakhatrana	2002	2005	2002	2002
Rapar	2009	2002	2002	2009

they intersect at very early/late period, i.e. 1982 for Bhuj (Figure 2(d)), 1983 for Mandvi (Figure 2(f)), 2011 for Mundra (Figure 2(g)). The presence of the CPs may not be considered adequate at very early/late years of the study period because of the uncertainty associated with lack of the preceding/following years. The beginning of the CPs at stations Naliya, Dayapar and Nakhatrana occurs in the years 1987, 1989 and 1987, respectively, as revealed by the existence of the intersection point within the confidence intervals at both 0.05 and 0.01 l.s.

From the above discussion, it can be inferred that the approach of using multiple tests for exploring the CPs is very much useful to avoid the straightforward and erroneous results. Also, it is revealed that the results of the sequential M–K test do not agree with those obtained from the SNH, Pettitt, CD and Bayesian tests. In contrast, the results of the SNH and Bayesian tests perfectly matched to each other.

3.2. Comparative evaluation of CP tests

All the expected CPs detected earlier were evaluated for their significance by applying one parametric t -test and one nonparametric M–W test, and the results are summarized in Table 2. All the expected CPs detected by the four tests for every station were evaluated to test for their significance at l.s. (α) of 0.05 and 0.10. When two CPs were found to be significant for a station then the one with high value of the test statistics was considered as the CP. The significance of a CP detected in the initial or last 5-year period, i.e. during 1980–1985 or 2009–2013, could not be examined due to non-availability of the sufficient data for further confirmation. A CP was considered to be the significant only after its confirmation by at least one of the employed evaluation tests.

In this study, the SNH and Bayesian tests identified the abrupt changes towards the tail end of the rainfall series, i.e. 2009 for Bhachau and Rapar stations, and 2012 for Mundra station. Also, the sequential M–K test detected the CPs either at the beginning years (1982 for Bhuj and 1983 for Mandvi) or at the ending year (2011 for Mundra). This tendency of the SNH and Bayesian tests to detect the CPs from the beginning or end years of the series is also reported in the earlier studies (Martínez

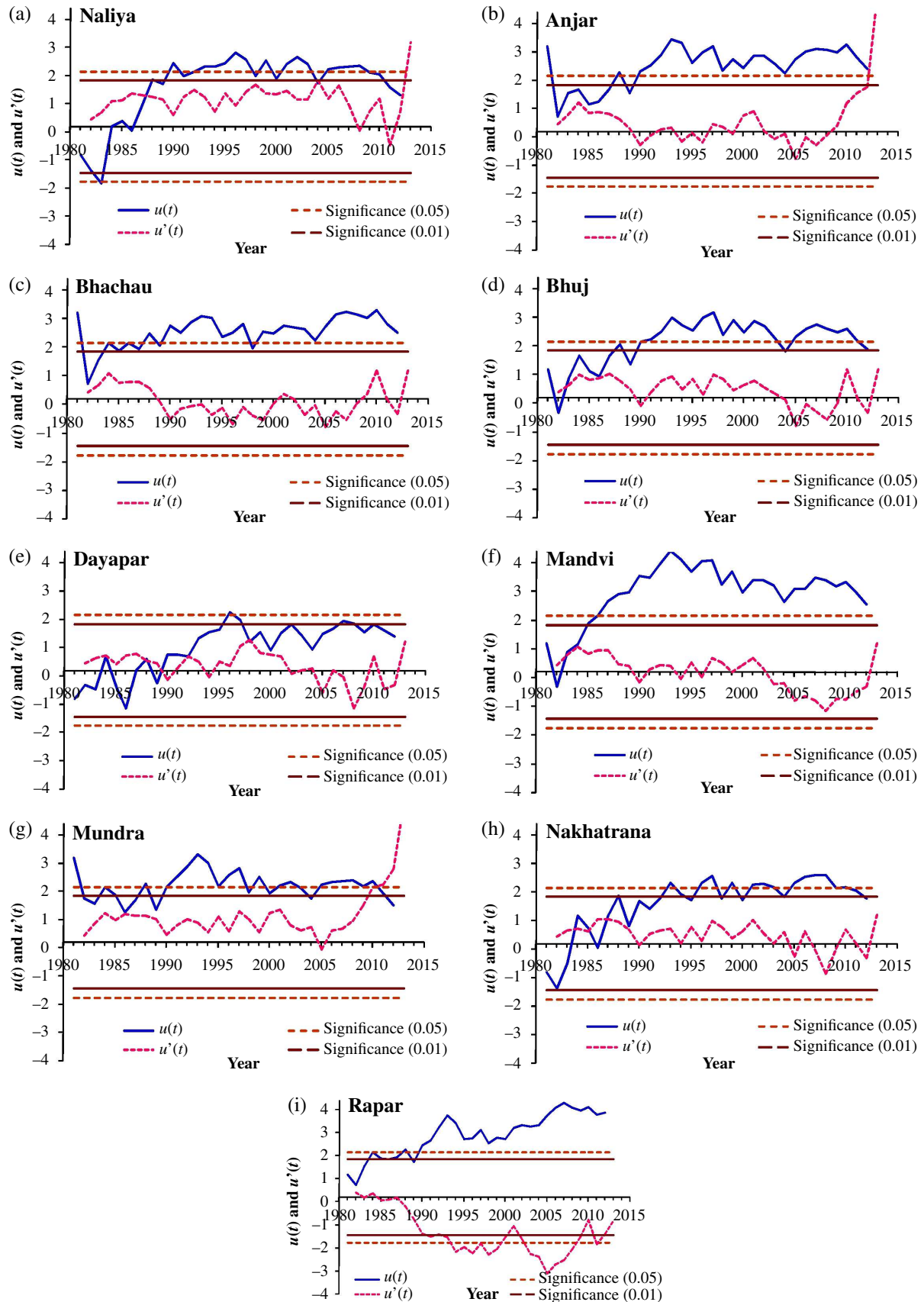


Figure 2. (a)–(i) Forward and backward sequence, $u(t_j)$ and $u'(t_j)$ of the normalized variable statistic of the Mann–Kendall test for nine stations.

Table 2. Observed test statistic values of t -test and Mann–Whitney test confirming significance of the identified change points in annual rainfall.

Rainfall station	Change point	t_s	u_c
Naliya	2005	1.72*	−1.74**
Anjar	2002	1.76*	−1.84**
Bhachau	1993	1.79*	−1.33
	2002	1.85*	−1.62
Bhuj	2002	1.70*	−1.59
	2005	1.95*	−1.89**
Dayapar	1987	0.84	−0.40
	1991	0.83	−0.42
	2002	0.42	−0.97
Mandvi	2005	2.46*	−2.36*
Mundra	2002	0.97	−1.23
Nakhatrana	2002	1.17	−1.33
	2005	1.15	−1.62
Rapar	2002	3.26*	−2.99*

t_s is test statistic of t -test; u_c is test statistic of Mann–Whitney test; boldface figures indicate significant or probable change points. Presence of significant change point at * $\alpha = 0.05$; ** $\alpha = 0.10$.

et al., 2009; Machiwal and Jha, 2016). Therefore, this study emphasizes the approach of using multiple tests for identifying the abrupt changes reasonably in hydrologic time series.

It is seen from Table 2 that there was single expected CP at five stations, i.e. year 2005 at Naliya, 2002 at Anjar, 2005 at Mandvi, 2002 at Mundra, and year 2002 at Rapar. The remaining stations were having either two CPs, i.e. years 1993 and 2002 for Bhachau, years 2002 and 2005 for Bhuj and Nakhatrana, or three CPs at a single station, i.e. years 1987, 1991 and 2002 for Dayapar. It is observed that the single CPs of the four stations, i.e. Naliya, Anjar, Mandvi and Rapar were confirmed by the t -test at $\alpha = 0.05$. The significance of the CPs at these four stations was also confirmed by the M–W test at $\alpha = 0.05$ (Mandvi and Rapar) and 0.10 (Naliya and Anjar). The single CP at Mundra station in the year 2002 could not be confirmed to be significant by any of the tests. Both the CPs detected at Bhachau station were confirmed to be significant by the t -test at $\alpha = 0.05$ but none of them could be confirmed by the M–W test. Furthermore, looking at the high value of t_s test statistic, the year 2002 was considered as the significant CP for the annual rainfall series of Bhachau station. Similarly, both the CPs of Bhuj station were confirmed to be the significant by the results of t -test at $\alpha = 0.05$; year 2005 was also confirmed by the M–W test at $\alpha = 0.10$. Hence, the annual rainfall of Bhuj was considered to have a significant and abrupt change in the year 2005. In contrast, both the CPs in the annual rainfall of Nakhatrana station could not be found significant by any of the tests at any of the α l.s. Likewise, none of the three expected CPs in the annual rainfall of Dayapar station could be revealed as the significant by the two tests at α of 0.05 and 0.10. However, the year 2005 may be considered as the CP for the annual rainfall of Nakhatrana station, and the year 2002 may be a probable CP for Dayapar station as the value of u_c test statistic

of the M–W test is the highest among the expected CPs (Table 2).

It is evident from the above results and their discussion that the year 2002 is confirmed to be the significant CP for the annual rainfall of Anjar, Bhachau and Rapar stations, and is a probable CP for Dayapar and Mundra stations. However, the significance of the year 2005 is verified for the occurrence of the abrupt CP in the annual rainfall of Naliya, Bhuj and Mandvi stations, and is the likely CP for Nakhatrana station. Based on the findings of the evaluation tests, it is clearly revealed that the results of the Pettitt test are the superior followed by the CD test over the other CP identification tests. However, the sequential M–K test failed in identifying the adequate CPs in the annual rainfall. The better accuracy of the Pettitt test in identifying CPs is also reported by Villarini *et al.* (2009) when compared with other four tests, i.e. cumulative sum, Guan, Rodionov and Bayesian methods. Relatively good performance of the Pettitt test may be attributed to its robust nature towards the presence of outliers and skewed distribution of the data series (Wijngaard *et al.*, 2003; Sagarika *et al.*, 2014). This kind of rank-based and distribution-free statistical test is recommended while dealing with hydrological data, which are often strongly skewed (Kundzewicz and Robson, 2004).

This finding reveals that the CPs in the annual rainfall of the study area occurred in 2 years, i.e. 2002 and 2005. This indicates that the annual rainfall of the arid western region of the country has significantly changed after the year 2001, which may be due to one or more possible factors, e.g. climate variability (Vivekanandan *et al.*, 2008; GEC, 2011), increased intensity of cyclones hitting the western Indian coastal land (GEC, 2011), increase in sea surface temperature (Rao and Vivekanandan, 2008), occurrence of the major earthquake in January 2001 in the area (Trivedi *et al.*, 2014), and/or cosmic influence on the sun–earth environment (Mukherjee, 2008). Moreover, future studies may be undertaken for detailed investigations to find the actual causes behind these abrupt rainfall changes.

3.3. Influence of CPs on annual rainfall

The influence of the identified CPs on the annual rainfall of nine stations was understood by computing three statistical properties, i.e. mean, SD and CV, for the complete rainfall series and sub-series before/after the confirmed CP, and the same are mentioned in Table 3. The mean annual rainfall in the area varies from 323 to 479 mm over the entire 34-year (1980–2013) period. It is observed from Table 3 that the mean of the rainfall sub-series after the CP is higher than that for the sub-series before the CP at all the stations. Before the CPs, the mean annual rainfall ranged from 306 (Bhuj) to 444 mm (Mundra); however, after the CP, the mean rainfall varied from 350 (Dayapar) to 627 mm (Mandvi) with the lowest increase of 14% (Dayapar) to the highest increase of 80% (Rapar). The SD of the annual rainfall for the entire time series varied from 238 (Anjar) to 340 mm (Mandvi). Among the nine stations, the SD of the annual rainfall at six stations increased after the CP with

increment of 8 (Naliya) to 173 mm (Mandvi); the SD of the annual rainfall decreased at three stations with a decrement varying from 59 (Mundra) to 101 mm (Nakhatrana). The CV values of the annual rainfall for the 34-year period ranged from 60 (Mundra) to 85% (Dayapar). It is further seen that the CV values of the annual rainfall after the CP are less than that before the CP at all the stations (Table 3). The reduction in the CV values after the CP ranges from 7 (Bhachau) to 42% (Nakhatrana). This finding indicates that the distribution of the annual rainfall after the CP becomes more uniform over the years with slightly less deviation from the mean values.

Furthermore, the box-whisker plots of the annual rainfall for nine stations were plotted for the two sub-series of the entire rainfall series bifurcated at the CP, and the same are depicted in Figure 3. It is worth mentioning that the second sub-series started with the CP. It is seen from Figure 3 that the median values of the annual rainfall are higher after the CP compared to those before the CP at all the stations. Also, the lower whiskers for all the stations got shifted at the higher level after the CP in comparison to that before the CP. This indicates that there was not a single year when negligible rainfall occurred in a year in the study area. It is observed that the length of the box little expanded after the CP at seven stations, except Mandvi and Mundra where the box-length showed a decrease. Besides, it is observed that the length of the upper whisker is somewhat shrunk after the CP for seven stations, i.e. Naliya, Anjar, Bhachau, Bhuj, Mandvi, Mundra and Nakhatrana. An important observation of the box-plots of the annual rainfall is that there are relatively fewer outliers/extremes in the annual rainfall series after the CPs (Figure 3). These findings clearly suggest that the magnitude of the annual rainfall has increased after the CPs at all the stations, and the distribution of the annual rainfall after the CPs is more uniform over the years at most of the stations.

3.4. Gradual trends before/after CPs

The gradual trends in the annual rainfall were identified for the entire 34-year (1980–2013) period, and for the two sub-series before and after the CPs, by applying three nonparametric tests, and their results are summarized in Table 4. The results of the SROC test did not reveal any significant trends in the annual rainfall series before and after the CPs at all the stations except at Bhachau station where a significant trend is revealed at $\alpha = 0.10$ l.s. However, in the 34-year period series, the SROC test indicated the significant trends of the annual rainfalls at four stations, i.e. Rapar ($\alpha = 0.01$), Bhachau ($\alpha = 0.05$), Anjar ($\alpha = 0.10$) and Mandvi ($\alpha = 0.10$).

The results of the KRC test revealed the negative trends in the annual rainfall before the CPs at seven of total nine stations (Table 4). The positive rainfall trends were present only at Bhachau and Rapar stations. However, none of the positive or negative trends were found to be statistically significant at any l.s. Of the seven stations having negative trends in the rainfall sub-series before the CPs, five stations revealed positive trends after the CPs;

positive trends in both the sub-series, i.e. before and after CPs, were obtained for Bhachau and Rapar stations, and the negative trends were found to be present in both the before and after sub-series of Naliya and Dayapar stations. In case of the entire 34-year rainfall series, all the stations were having the positive (or increasing) rainfall trends as revealed from the results of the KRC test (Table 4); the statistically significant trends were observed only at four stations, i.e. Rapar ($\alpha = 0.01$), Anjar ($\alpha = 0.10$), Bhachau ($\alpha = 0.10$) and Mandvi ($\alpha = 0.10$).

In the annual rainfall sub-series before the CP, based on the results of the M–K test, only two stations (Bhachau and Rapar) revealed positive rainfall trends, and the rest seven stations showed declining trends of the annual rainfall (Table 4). Whereas, in the rainfall sub-series after the CP, the decreasing trends could be observed only at three stations, i.e. Naliya, Dayapar and Nakhatrana. Similar to the results of the KRC tests, none of the positive or negative rainfall trends obtained from the M–K test was found to be statistically significant at any l.s. However, in case of the complete 34-year period rainfall series, the statistically significant inclining trends of the annual rainfall were present at four stations, i.e. Rapar ($\alpha = 0.05$), Anjar ($\alpha = 0.10$), Bhachau ($\alpha = 0.10$) and Mandvi ($\alpha = 0.10$) (Table 4).

It is obvious from the above interpretations that the results of the three nonparametric trend identification tests are in good coherence to each other for the 34-year annual rainfall series of all nine stations. Thus, the obtained results indicating presence of the significant rainfall trends at four stations are quite strong and precise. This finding indicates that the rainfall in the study area is considerable rising. It is apparent that the gradual trends are generally visible over a relatively long duration of time such as the 34-year period of the entire rainfall series in this study. The presence of the gradual trends could not be dominating in either of two short duration rainfall sub-series, before and after the CP.

3.5. Magnitudes of gradual rainfall trends

The magnitudes of the increasing/decreasing trends were estimated by using linear regression modeling, by fitting linear regression lines to the two rainfall sub-series before and after the CP, and the results are presented in Figure 4. It is well discernible from Figure 4 that the annual rainfall before the CP had decreasing rainfall trends at five stations with magnitudes between 0.9 (Anjar) and 6.2 mm year⁻¹ (Mundra). The increasing trends of the annual rainfall before the CP could be seen at four stations with magnitudes varying from 1.6 (Nakhatrana) to 7.1 mm year⁻¹ (Rapar). The sub-series of the annual rainfall after the CPs revealed the increasing trends at eight stations with the smallest magnitude of 4.7 mm year⁻¹ (Nakhatrana) and the largest magnitude of 40.8 mm year⁻¹ (Bhachau); Dayapar is the only station that showed a decreasing rainfall trend after the CP with a magnitude of 7.3 mm year⁻¹. An important finding of this section is that the values of the trend magnitudes for the second rainfall sub-series after the CPs are comparatively larger and more positive than those for

Table 3. Basic statistical properties of the rainfall time series at nine stations for period 1980–2013 and for the periods before and after change points.

Statistical property	Period	Naliya	Anjar	Bhachau	Bhuj	Dayapar	Mandvi	Mundra	Nakhatrana	Rapar
Mean (mm)	Entire period	379.5	384.4	400.0	353.4	322.7	405.2	479.0	380.7	407.4
	Before change	337.7	333.0	341.7	306.2	307.9	325.2	443.7	350.0	317.5
	After change	495.6	478.7	506.8	484.8	349.8	627.6	543.6	466.1	572.3
Standard deviation (mm)	Entire period	243.2	237.9	258.1	246.0	275.6	340.1	287.9	261.9	247.7
	Before change	234.3	226.5	223.6	227.1	303.2	264.5	307.3	282.4	187.8
	After change	242.1	238.3	291.8	261.6	226.1	437.6	247.9	180.8	266.3
Coefficient of variation (%)	Entire period	64	62	65	70	85	84	60	69	61
	Before change	69	68	65	74	98	81	69	81	59
	After change	49	50	58	54	65	70	46	39	47

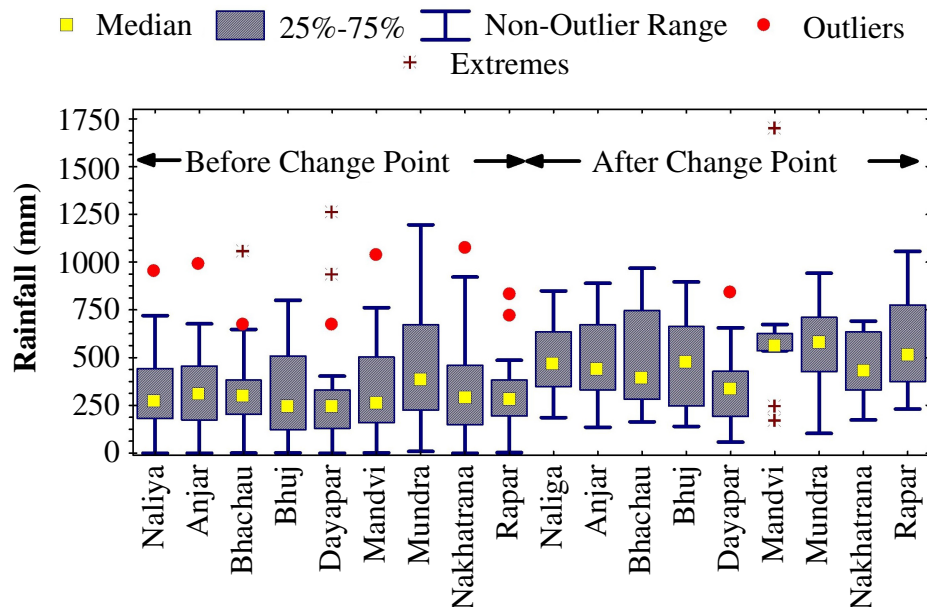


Figure 3. Box-whisker plots of the annual rainfall for nine stations before and after the change points.

Table 4. Calculated and critical test statistic values of trend tests for the annual rainfall time series before and after change points.

Rainfall station	Spearman rank-order correlation test			Kendall rank correlation test			Modified Mann–Kendall test		
	Before CP	After CP	Entire period	Before CP	After CP	Entire period	Before CP	After CP	Entire period
Naliya	0.94	0.18	0.72	−1.07	−0.21	0.58	−1.35	−0.28	0.63
Anjar	0.16	1.13	1.62*	−0.25	1.10	1.92*	−0.26	1.43	1.84*
Bhachau	0.21	1.58*	1.74**	0.14	1.51	1.73*	0.14	1.64	1.87*
Bhuj	0.51	0.22	1.11	−0.56	0.42	1.11	−0.59	0.33	1.13
Dayapar	0.29	0.07	0.77	−0.14	−0.27	0.67	−0.14	−0.49	0.82
Mandvi	0.30	1.15	1.62*	−0.33	1.25	1.76*	−0.42	1.31	1.70*
Mundra	0.71	0.92	0.79	−0.87	0.89	0.76	−1.21	1.33	1.03
Nakhatrana	0.32	0.22	1.13	−0.28	0.00	1.02	−0.34	−0.14	1.39
Rapar	0.70	1.26	3.45***	0.65	1.23	3.04***	0.56	1.20	2.38**

Presence of significant change point at * $\alpha = 0.10$; ** $\alpha = 0.05$; *** $\alpha = 0.01$.

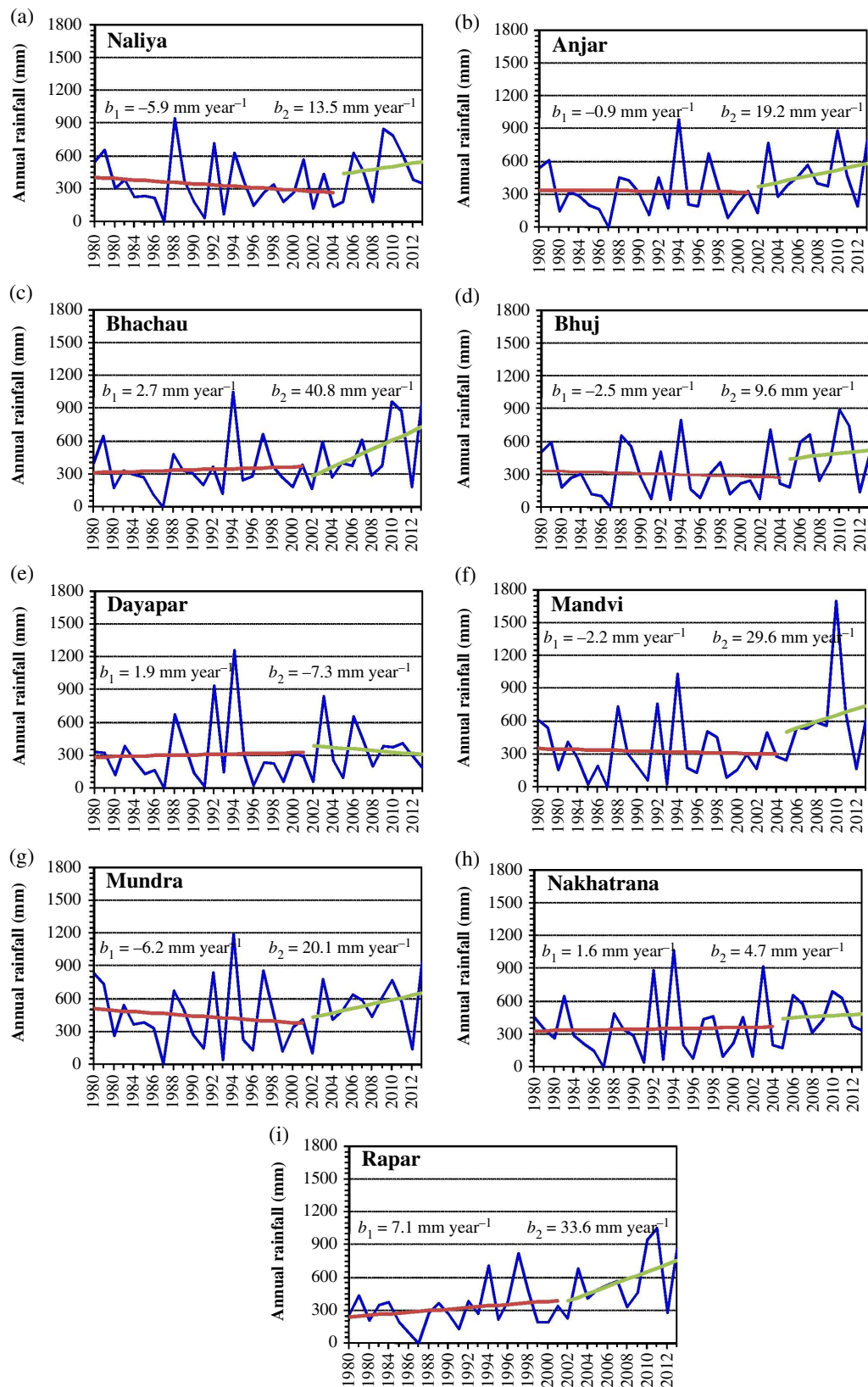


Figure 4. (a)–(i) Magnitudes of the increasing/decreasing trends estimated by using linear regression modeling, along with linear regression lines fitted separately to rainfall sub-series before and after change points.

Table 5. Magnitudes of increasing/decreasing annual rainfall trends for periods 1980–2013 and before/after change points.

Station	Entire period		Trend magnitudes (mm year ⁻¹)		After change point	
	Q_{med}	b	Q_{med}	b	Q_{med}	b
Naliya	3.00	3.33	-6.58	-5.90	-2.27	13.49
Anjar	6.00	6.48	-3.63	-0.88	23.66	19.18
Bhachau	5.25	9.17	0.33	2.69	52.09	40.77
Bhuj	5.07	5.31	-3.46	-2.52	19.72	9.59
Dayapar	2.64	1.91	-1.35	1.99	-7.04	-7.35
Mandvi	9.00	10.07	-1.73	-2.18	20.01	29.64
Mundra	4.39	3.22	-11.89	-6.24	32.63	20.11
Nakhatrana	5.00	4.71	-1.58	1.59	-1.00	4.65
Rapar	11.38	13.68	3.33	7.14	43.64	33.63

Q_{med} is trend magnitude by Sen's slope estimator; b is trend magnitude by slope of linear regression.

the first sub-series before the CPs. This clearly highlights that the trend magnitudes in the annual rainfall of the study area are rising regardless of their statistical significance.

Furthermore, the nonparametric Sen's slope estimation test was used to quantify the magnitudes of the trends in the entire 34-year period series and two sub-series of the annual rainfall of nine stations, and the results are given in Table 5. It is apparent from this table that the annual rainfall before the CPs decreased at seven of the total nine stations at a rate of 1.35–11.89 mm year⁻¹; however, after the CP, the annual rainfall increased at six stations at a rate of 19.72–52.09 mm year⁻¹. This indicated that the magnitude of the positive trends of the annual rainfall considerably increased after the CP. When the entire 34-year annual rainfall series was considered, all the stations revealed the increasing rainfall rates ranging from 2.64 mm year⁻¹ (Dayapar) to 11.38 mm year⁻¹ (Rapar). Thus, the magnitudes of the annual rainfall trends after CPs were substantially high ranging from 2.22 to 9.92 times more than that for the entire period at six stations, i.e. Anjar, Bhachau, Bhuj, Mandvi, Mundra and Rapar (Table 5). The magnitudes of the increasing rainfall trends at most of these six stations are the highest for the period after CPs in comparison to those for the periods before CPs and of the entire 34-year period. It is also seen that the results of the Sen's slope estimation test showed a good harmony with the results of the linear regression test.

3.6. Significance of gradual trends over sequential years

In this study, a new approach is adopted by applying the M–K test to sequential rainfall series for investigating the effect of increasing number of years on the significance of gradual trends of annual rainfall. The results of this new approach, illustrated by bar charts of the z -statistic values of the M–K test, fitted linear regression line to the z -statistic bar charts along with significance lines at $\alpha = 0.05$ and 0.10 for the annual rainfall series of the nine stations, are presented in Figure 5(a)–(i). It can be clearly seen from Figure 5(a)–(i) that the bar charts of the z -statistic values for eight stations are extending downward from the x -axis indicating negative z -statistic values during the initial few periods, which continuously go on

approaching the x -axis, and then start inclining upward on the x -axis indicating a continuous increment in the positive values of the z -statistics. However, for Rapar station, the z -statistic values are found to be the positive throughout all the periods, and the bar charts are upward during the initial periods also.

It is further observed from Figure 5(a)–(i) that the linear regression lines are perfectly fitted to the bar charts of the z -statistic values for all the stations as revealed from the high R^2 values ranging from 0.78 (Naliya) to 0.96 (Rapar). Depending upon the R^2 values, the linear trends of the annual rainfall ascertained by the z -statistic values may be considered of highly strong nature ($R^2 \geq 0.90$) for five stations, i.e. Anjar, Bhachau, Mandvi, Mundra and Rapar. However, the linear trends are moderately strong ($0.90 > R^2 \geq 0.75$) for four stations, i.e. Naliya, Bhuj, Dayapar and Nakhatrana. This finding suggests that the z -statistic value for the annual rainfall time series of all the stations have been continuously increasing after the year 2000, which is closer to the CPs, i.e. years 2002 and 2005. With the continuous increase in the positive z -statistic values over the sequential periods, the significance of the linear trends in the annual rainfalls of the stations also increases. The rainfall trends in the year 2013 already became statistically significant at Anjar, Bhachau and Mandvi stations at $\alpha = 0.10$, and at Rapar station at $\alpha = 0.01$ as the fitted linear regression lines at these station cross the significance lines (Figure 5(b), (c), (f) and (i)). Nevertheless, at other stations, the fitted linear curves may cross the significance lines in the coming years, and hence, the linear trends of the annual rainfall may be found statistically significant in near future as inferred from the strong to very strong nature of the increasing trends of the z -statistic values. In addition, for the stations where the rainfall trends are already statistically significant, the significance levels of the trends may also be increased from $\alpha = 0.10$ to 0.05 and then to 0.01.

The above results clearly explain that the significance of the gradual trends in the annual rainfall of the study area is continuously increasing from the chosen base period, i.e. 1980–2000. Therefore, the gradual rainfall trends may be statistically more significant in the coming years under the scenario of the global climate change and variability.

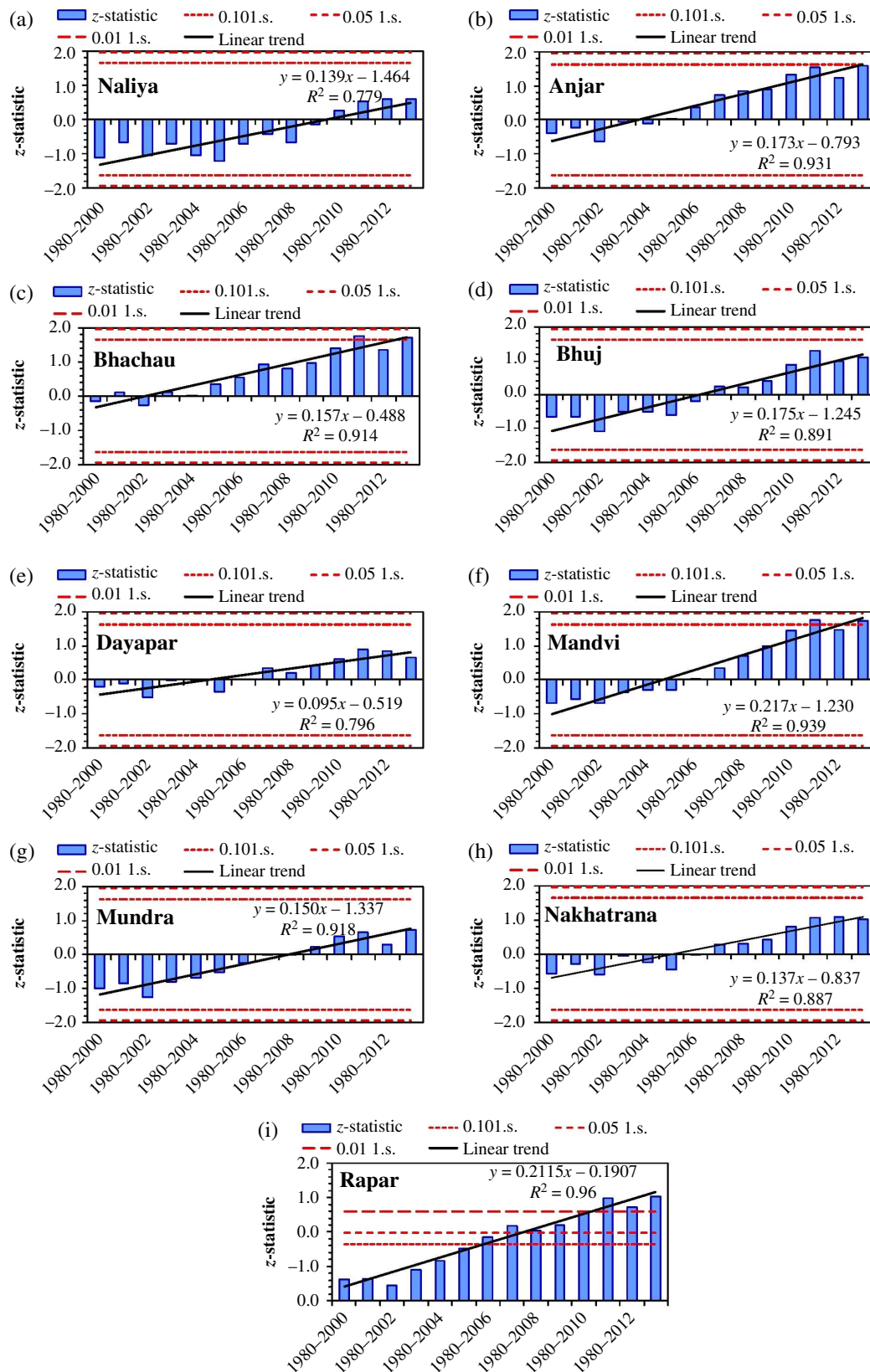


Figure 5. (a)–(i) Bar charts of Mann–Kendall test statistic values for sequential annual rainfall series starting from 1980–2000 up to 1980–2013, along with fitted linear regression lines, critical limits at three level of significance (1.s.), coefficient of variation (R^2) values and fitted linear equation.

4. Conclusions

This study examined presence of gradual trends and abrupt changes in annual rainfall series of nine rainfall stations in an Indian arid region. The trends and CPs were statistically quantified and verified, and need of adopting multiple statistical tests is justified. Results of SNH test and Bayesian test closely matched to each other with tendency to identify the CPs from beginning or end of the series. Performance of Pettitt test was found to be superior followed by CD test, among others. Furthermore, *t*-test and M–W test confirmed the years 2002 and 2005 as the CPs at five and four stations, respectively. The abrupt changes increased the mean annual rainfall by 14–80% with 7–42% reduction in values of coefficient of variation. Box–whisker plots further indicated that shift in rainfall regime at the CPs made the rainfall distribution more uniform over the years. In 34-year rainfall, results of SROC, KRC and M–K tests detected significant trends at four stations, i.e. Anjar, Bhachau, Mandvi and Rapar at l.s. of 0.01 or 0.05. However, in rainfall sub-series before/after the CP, no test could reveal significant trends except at Bhachau station where significant trend was found by SROC test. Linear regression analysis revealed relatively large trend magnitudes for the rainfall sub-series after the CP. Also, Sen's slope test showed that magnitudes of the most rainfall trends after the CPs were 2–10 times higher than that for entire 34-year period. These findings clearly illustrate that the rainfall in the study area is considerably rising. The novel approach of this study ascertained strong linear trends ($R^2 = 0.78\text{--}0.98$) of M–K test statistic values over sequential years, which suggests that the annual rainfall may further increase in near future along with increased l.s.. Moreover, the above findings are useful for policy makers and planners to develop suitable strategies to cope up with heightened rainfall rise. The methodology and results presented here may be employed in other climate vulnerable regions of the world.

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